

V. TOPOLOGY AND MONITORED DYNAMICAL PHASES—ADDITIONS

41a. A. Lavasani, Y. Alavirad, and M. Barkeshli, “Measurement-Induced Topological Entanglement Transitions in Symmetric Random Quantum Circuits,” *Nature Physics* **17**, 342–347 (2021). [NEW] [[arXiv](#)] [[DOI](#)]

Summary. Competing measurements and symmetry-preserving random unitaries produce distinct symmetry-protected topological and trivial area-law phases, together with a volume-law phase. In the measurement-only limit, the transition maps to two copies of two-dimensional percolation; sparse unitaries split this tricritical point into two critical lines. The work is a foundational example in which topology belongs to the ensemble of monitored trajectories and is visible in trajectory entanglement rather than ordinary linear expectation values.

41b. A. Lavasani, Y. Alavirad, and M. Barkeshli, “Topological Order and Criticality in $(2 + 1)$ D Monitored Random Quantum Circuits,” *Phys. Rev. Lett.* **127**, 235701 (2021). [NEW] [[arXiv](#)] [[DOI](#)]

Summary. Random Clifford unitaries, single-qubit measurements, and toric-code stabilizer measurements generate topologically ordered, trivial area-law, volume-law, and critical trajectory phases in two spatial dimensions. The phase diagram contains a percolation tricritical point and a distinct anisotropic self-dual tricritical point. This is the direct higher-dimensional monitored-circuit precedent for asking how a spatially patterned dynamical phase terminates at an interface.